

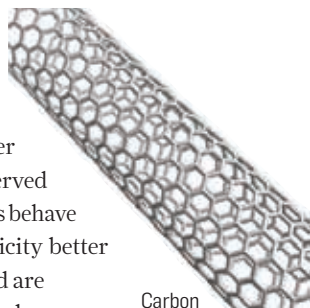
providing efficient biological markers, and advance the development of optical devices such as light emitting diodes (LEDs).

Manipulation of Atoms (1990)

Using a Scanning Tunneling Microscope (STM), IBM researchers Donald Eigler and Erhard Schweizer were able to arranged individual Xe atoms on a surface. At IBM's Almaden Research Center in San Jose, California, they spelled out the company logo using 35 xenon atoms, which now hangs in IBM's STM Image Gallery.

Carbon Nanotubes (1991)

Sumio Iijima of NEC is credited with discovering the carbon nanotube (CNT), although there were early observations of tubular carbon structures (several tubes nested inside each other) by others as well. Two years later Iijima, Donald Bethune at IBM in the US and others observed single-walled nanotubes just 1-2 nanos in dia. Nanotubes behave like metals or semiconductors, but can conduct electricity better than copper, can transmit heat better than diamond, and are among the strongest materials known. Nanotubes could play a pivotal role in the practical applications of nanotechnology if their remarkable electrical and mechanical properties can be exploited.



Carbon
Nanotubes



Mounji Bawendi of MIT

Synthesis of nano-Crystals (1993)

Mounji Bawendi of MIT invented a method for controlled synthesis of nanocrystals (quantum dots), paving the way for applications ranging from computing to biology to high-efficiency photovoltaic and lighting. Within the next several years, work by other researchers such as Louis Brus and Chris Murray also contributed methods for synthesising quantum dots.

IWGN (1998)

The Interagency Working Group on Nanotechnology (IWGN) was formed under the National Science and Technology Council to investigate the state of the art in nanoscale science and technology and to forecast possible future developments. The IWGN's study and report, "Nanotechnology Research Directions: Vision for the Next Decade" defined the vision for